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Prediction of micropitting damage in gear teeth contacts considering the concurrent effects of surface fatigue and mild wear

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ABSTRACT

The present paper studies the occurrence of micropitting damage in gear teeth contacts. An existing general micropitting model, which accounts for mixed lubrication conditions, stress history, and fatigue damage accumulation, is adapted here to deal with transient contact conditions that exist during meshing of gear teeth. The model considers the concurrent effects of surface fatigue and mild wear on the evolution of tooth surface roughness and therefore captures the complexities of damage accumulation on tooth flanks in a more realistic manner than hitherto possible. Applicability of the model to gear contact conditions is first confirmed by comparing its predictions to relevant experiments carried out on a triple-disc contact fatigue rig. Application of the model to a pair of meshing spur gears shows that under low specific oil film thickness conditions, the continuous competition between surface fatigue and mild wear determines the overall level as well as the distribution of micropitting damage along the tooth flanks. The outcome of this competition in terms of the final damage level is dependent on contact sliding speed, pressure and specific film thickness. In general, with no surface wear, micropitting damage increases with decreasing film thickness as may be expected, but when some wear is present micropitting damage may reduce as film thickness is lowered to the point where wear takes over and removes the asperity peaks and hence reduces asperity interactions. Similarly, when wear is negligible, increased sliding can increase the level of micropitting by increasing the number of asperity stress cycles, but when wear is present, an increase in sliding may lead to a reduction in micropitting due to faster removal of asperity peaks. The results suggest that an ideal situation in terms of surface damage prevention is that in which some mild wear at the start of gear pair operation adequately wears-in the tooth surfaces, thus reducing subsequent micropitting, followed by zero or negligible wear for the rest of the gear pair life. The complexities of the interaction between the contact conditions, wear and surface fatigue, as evident in the present results, mean that a full treatment of gear micropitting requires a numerical model along the lines of that applied here, and that use of overly simplified criteria may lead to misleading predictions.

1. Introduction

Micropitting is an important failure mechanism in modern day machine elements employing heavily-loaded, sliding-rolling, lubricated contacts, such as gears and rolling element bearings. Micropitting is caused by rolling contact fatigue which occurs due to repeated over-rolling and hence cyclic contact stresses. It differs from pitting in that the damage occurs at surface asperity level rather than the macro contact level. As such, micropitting is prevalent in contacts operating under low specific film thicknesses (low Λ ratios) and mixed lubrication regime, typical of gear teeth contacts. With the current trend of decreasing lubricant viscosities and increasing power densities of mechanical systems, with the ultimate aim of increasing mechanical efficiency, lubricant films protecting operating surfaces are getting thinner leading to increased metal to metal contact on roughness asperity level and consequently, an increasing incidence of micropitting. Despite its importance, there are currently no accepted gear design criteria for prevention of micropitting and the subject still suffers from the lack of full understanding. The present paper attempts to address some of the present deficiencies by firstly, proposing a relatively simple predictive tool for the onset of micropitting in gear teeth contacts and secondly, using this model to gain further insight into the influence of relevant factors on the onset of micropitting, with a particular emphasis on the role of slide-roll-ratio in gear teeth contacts and the effects of

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Nomenclature

A	Wöhler curve slope parameter [Pa]
B	Wöhler curve intercept parameter [Pa]
A_p	Micropitted area proportion [–] or [%]
D	Dawson's roughness/film thickness ratio [–]
h	Film thickness value [m]
h_c	Central film thickness value [m]
k_{lub}	Dimensional wear coefficient [s]
m_n	Gear tooth module [mm]
p	Contact pressure [Pa]
P	Pitch line location [–]
Δp	Hydrodynamic pressure fluctuation [Pa]
p_o	Maximum Hertzian pressure [Pa]
U	Dowson-Higginson speed parameter [–]
u	Surface speed [m/s]
$\bar{u}$	Average (entrainment) speed of the surfaces, $\bar{u} = (u_1 + u_2)/2$
R_q	Root-mean-square (RMS) roughness parameter [m]
S	Sliding/rolling ratio, $S = (u_1 - u_2)/\bar{u}$ [–]
W	Dowson-Higginson load parameter [–]
t	Time [s]
x	Co-ordinate, rolling direction [m]
$x_{1,2}$	Profile shift in the gear involute, gear 1 and 2 [–]
y	Co-ordinate, transverse-to-rolling direction [m]
z	Co-ordinate, normal-to-contact direction [m]
$z_{1,2}$	Number of gear teeth, gear 1 and 2 [–]
Δt	Time period [s]
Λ	Specific film thickness or Lambda ratio, $\Lambda = h_c/R_q$ [–]
ω	Rotational speed [rad/s]
φ	Roll angle during gear meshing [deg]
τ_{VM}	von Mises stress [Pa]

mild wear.

Micropitting in lubricated Hertzian contacts has been studied since 1930s. The pioneering work of Way [1] describes micropitting and its relation to the surface finish. Dawson [2–4], with the help of a two-disc experimental set-up, first related micropitting to a ratio D, equal to the reciprocal of what is now known as the specific film thickness or Λ–ratio, i.e. D = total initial surface roughness of the two discs / oil film thickness.

Largely owing to the work of Olver and co-workers [5–8], micropitting is today recognized to be a surface contact fatigue phenomenon that involves a competition between asperity level fatigue and mild wear. Asperity interaction in poorly lubricated rolling-sliding, heavily-loaded contacts produces surface fatigue, while mild wear reduces the asperity heights and removes the top fatigued layers of material, which in turn acts to prevent or, at least reduce, the progression of surface fatigue. The competition between these two processes determines whether or not micropitting will develop in the first instance, and if it does, whether or not it will increase in severity with time ultimately leading to significant loss of tooth profile and gear failure. The first complete micropitting model addressing this competition and hence providing accurate prediction of micropitting progression was developed by Morales-Espejel and Brizmer [9], who initially applied it to the study of micropitting in rolling bearings. The same authors [10] later adapted this model to account for the effect of additives, which strongly influence wear and local boundary friction, on micropitting in rolling bearings.

Other numerical approaches for prediction of micropitting in gear teeth include those of Brandão et al. [11,12] and Li and Kahraman [13–15]. Evans et al. [16] use a mixed lubrication model to predict the micropitting damage accumulation during a single mesh cycle in a pair of helical gears. They use measured gear roughness and compare their results to experimental observations. The above models account for transient contact conditions along the path of contact and prevailing lubrication regime but omit the interaction between micropitting and wear. More recently Brandão et al. [17] extended their model to include mild wear but unlike in the approach of Morales-Espejel et. el. [9], as used in the present paper, the fatigue and wear damage are treated in succession rather than simultaneously so that their continuous mutual effects on each other are not accounted for.

Such predictive tools have only been made possible by experimental studies of micropitting which have served to provide significant understanding of the phenomenon. Oila and Bull [18] and Febre et al. [19] describe the relative influence of a range of factors affecting micropitting. Metalographical and morphological aspects of micropitting are described in [20–22]. Olver and co-workers [5–8] focus on the influence of lubricant additives and lambda ratio on the progression of micropitting using a triple-disc contact fatigue rig, while Rycerz and Kadiric [23] utilise the same rig to study the influence of slide-roll-ratio on micropitting.

Although, the model in [9] has been successfully used to predict micropitting in rolling bearings, occurrence and extent of micropitting damage is much more common in gears. This is primarily due to the higher surface roughness but also the adverse effects of any misalignment and tooth geometry deviations on prevailing gear tooth contact stresses [16]. Despite this, there is no universally accepted predictive tool for the onset of micropitting in gears and it therefore seems sensible to adapt this proven micropitting model to conditions pertinent to gear teeth contacts. Furthermore, there are currently ongoing efforts within the relevant standards organizations to provide design guidance for prevention of micropitting [24] and improved prediction of gear micropitting would serve to better inform the debate around the most acceptable design criteria. In particular, increased sliding has been suggested to lead to increased micropitting, with the proposed mechanism [24,25] being that higher sliding leads to higher contact temperatures and lower elasto-hydrodynamic film thicknesses, which in turn decreases the prevailing specific film thickness (Λ ratio) which is known to lead to higher micropitting. This proposed mechanism deserves further consideration, not least because slide/roll ratio is generally considered to have minimal effect on elasto-hydrodynamic film thickness, unless conditions close to pure sliding are encountered, and the fact that the occurrence of micropitting in rolling bearings, which operate at much lower levels of sliding with typical slide/roll ratios ($S = 2(u_2\text{-}u_1) / (u_1+u_2)$) being around 5%, has been demonstrated in literature [9].

This paper attempts to address some of the issues discussed here, by using a proven numerical model to study the influence of gear operating conditions on the onset of micropitting in real gear teeth contacts. The validity of the approach under contact conditions pertinent to gear teeth contacts is first confirmed by comparing the model predictions to experimental micropitting results obtained under controlled contact conditions on a triple-disc micropitting rig. Particular attention is paid to the role of sliding and the mechanism by which it may increase the extent of micropitting.

The model is then applied to a pair of meshing spur gears. Rather than simply providing global estimates of micropitting during the meshing cycle, the present approach attempts to provide local predictions of micropitting distribution along the tooth profile by considering the influence of local pressures, surface roughness, lubrication conditions and kinematics of a gear tooth contact during a meshing cycle, as well as the simultaneous effects of fatigue and wear on the local micro-contact geometry. Although experimental observations of micropitting distribution along the tooth flank exist in literature [16,26], the existing numerical treatments [11–17] lack detailed consideration of these local aspects during the meshing cycle. The existing micropitting model [9], previously used for prediction of micropitting in rolling bearings, is adapted here to spur gear teeth contacts and operating conditions, with an addition of an improved local wear model, described and validated in [27].

2. Research methodology

The general modelling methodology used in the present work is designed to simulate micropitting behaviour in any type of heavily loaded rolling-sliding lubricated contact, but has been successfully applied in the past to study the onset of micropitting in rolling element bearings [9]. For the purposes of the current study the model was adapted to gear applications, by making the necessary modifications to the relevant input parameters. These included material Wöhler curve parameters for gear steel, measured gear surface topography and accounting for the actual, transient contact conditions encountered along the contact path of meshing gears including the varying entrainment speed, slide/roll ratio and contact pressure. Detailed description and experimental validation of the model can be found in previous publications concerning the prediction of micropitting in rolling element bearings [9], therefore only a brief introduction to the model is given here.

2.1. Modelling method

The model calculates the severity of fatigue damage arising from Hertzian macro-cycles and roughness micro-cycles, including the effect of mild wear of surface asperities on roughness related stresses. The result is a continuously evolving damage map, showing the location of micro-pits on the contacting surfaces, thus allowing the micropitted area to be predicted as a function of simulated load cycles.

The model assumes homogenous, elastic-perfectly plastic material and mixed elastohydrodynamic lubrication conditions (EHL). The flowchart in Fig. 1 shows the general arrangement of the solver. First, operating conditions and lubricant rheology parameters are entered along with representative measured 3D topography maps of both contacting surfaces. A fast, isothermal non-Newtonian (Eyring) mixed lubrication model described in [28] is used to calculate the stress history per load cycle of each surface point by solving the problem at different time steps as the surfaces slide-roll against each other. This fast approach to modelling micro-elasto-hydrodynamic lubrication (micro-EHL) conditions determines the clearance between the mating surfaces and the pressure fluctuations arising from the passage of roughness asperities through the contact in mixed lubrication conditions. The smooth Hertzian pressure distribution is modelled separately and the overall pressure and clearance distributions in time during a load cycle are then predicted by superimposing the micro-EHL and Hertzian solutions. Friction coefficient is considered on the local level, so that any identified solid-to-solid contact areas within the mesh are assigned friction coefficient of 0.11, while areas where load is carried by a fluid film are assumed to have friction of 0.05. Dang Van fatigue criterion, together with Wöhler curve parameters for the given material, is used to determine the fatigue damage of each surface point for the given stress history. Since the surfaces, and hence the stresses, are continuously evolving over time due to micropitting and wear, some means of accounting for these transient conditions are required. The approach adapted here updates the surface roughness profiles every ΔN number of contact cycles. The actual ΔN number of cycles is chosen to give an appropriate compromise between calculation time and prediction accuracy given the overall number of contact cycles being simulated. The total accumulated damage is then accounted for by adding the fatigue damage acquired over each set of load cycles through application of Palmgren-Miner linear damage accumulation rule. The calculated fatigue damage is compared against a maximum allowable damage parameter and a material detachment model creates a micropit in the surface topography at the location where the parameter was exceeded as described in [9]. In addition, a local mild wear model, based on the Archard's wear law [27], with an appropriate Archard wear coefficient, is applied in parallel over the same set of cycles and the surface topography is updated accordingly. Theoretically, it is possible to apply the wear model, and Palmgren-Miner rule, for each load cycle, and update the surface topographies accordingly, but this would significantly increase the calculation effort while providing very little benefit in terms of accuracy of predictions, and is therefore deemed unnecessary, particularly in cases where there are no short lived, large shock loads and only mild wear is present. The contact solver then continues the calculation with the modified geometry, repeating the process described above until a desired number of total load cycles is reached. The model then returns the overall damage estimation in terms of micropitted surface area as well as a damage map showing the spatial distribution of micropitting.

In order to represent the fatigue behaviour of the gear steel in the simulations, the Wöhler curve parameters were estimated from the case-carburized low-carbon gear steel stress-life curves given in [29] for 10% probability of failure (L_{10}). The obtained parameters were $A = 40MPa$ and $B = 1600MPa$. In order to accurately predict mild wear of surface asperities, the micropitting model requires a wear coefficient as an input, the value of which is ideally established through a number of wear experiments with the given material under the relevant contact conditions. Since the subject of the present paper are general trends in micropitting behaviour of gear contacts, the exact value of wear coefficient for a particular application was not necessary and assumed

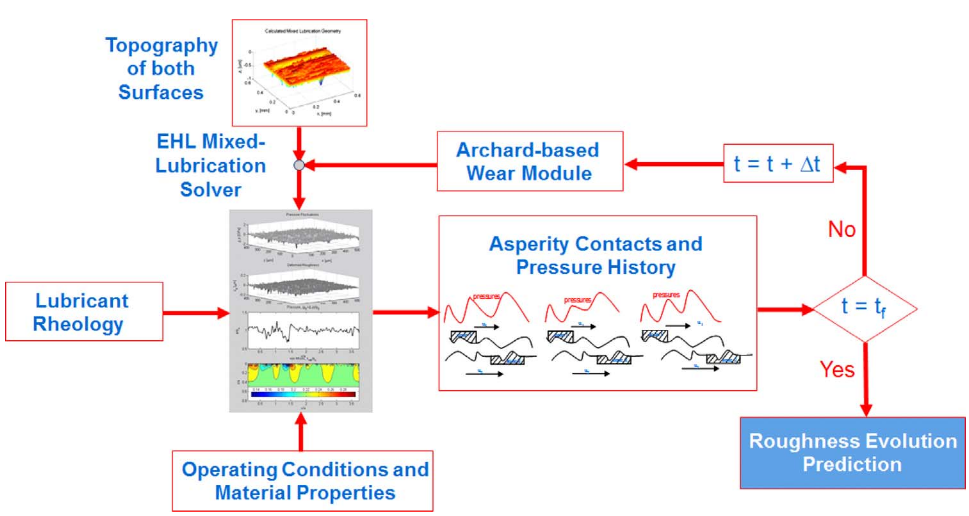


Fig. 1. Schematic of the algorithm used in the micropitting model illustrating the use of mixed lubrication model and concurrent treatment of wear and surface fatigue damage. Adapted from [9].

values are used. Indeed, the changes in micropitting behaviour under the same gear operating conditions but with different wear rates are of interest here, so that different wear coefficients were used in different sets of calculations, using the wear coefficient measured in previous rolling bearing experiments [9] as a reference to ensure that wear conditions remain realistic.

Unlike in rolling bearings, in gears, as the meshing cycle progresses, pressure distribution, contact geometry and sliding conditions vary, as illustrated schematically in Fig. 2a. The micropitting model described above is a transient model, in that the calculation is performed using the local contact conditions at a number of discrete locations along the gear meshing cycle. To account for the changing conditions, and obtain the associated micropitting distribution along the tooth flank, the model is applied at a number of selected locations on the tooth flanks covering the whole path of contact. At each of these locations, the simulation then considers the relative progression of the surface topographies of the mating teeth across the contact width. One contact cycle is the period for the surface roughness to fully transverse the contact width at a given location. Each of the contact cycles is discretized in several time steps, as illustrated graphically in Fig. 2b. The mesh size in all simulations was 131 × 131 × 15 elements in x, y, z directions respectively. 20 time steps were modelled to obtain stress history for each contact cycle, i.e. contact solver was applied 20 times between $t = 0$ and $t = t_f$, were t_f is the time needed for the roughness to cross the contact width in a given load cycle at a given location on the contact path.

2.2. Experimental set-up and procedure

The micropitting model used in this paper has previously been verified experimentally under conditions typical of rolling element bearings [9]. Before the model is applied to gear micropitting, its suitability for prediction of micropitting in gear applications is first confirmed by comparing its predictions to experimental measurements of micropitting obtained under conditions typical of gear teeth contacts. For this purpose, micropitting damage was generated under controlled conditions on a triple disc machine (PCS Instruments MPR rig), extensively used for micropitting studies in the past [6–9,23]. The rig is shown in Fig. 3a. The rig consists of three counterface discs in contact with a test roller sample, with the discs and the roller driven by independent motors, allowing any slide-roll ratio to be set. The roller specimen accumulates 3 contact cycles per revolution, thus this is the test specimen where the micropitting damage is preferentially generated. The test specimens were made of case hardened 16MnCr5 steel with a case depth of approximately 0.8 mm. The measured case hardness was 790HV and 680HV for the counterface discs and roller specimens respectively. The hardness differential is introduced in order to ensure that minimal wear to the counterface disc surfaces occurs during

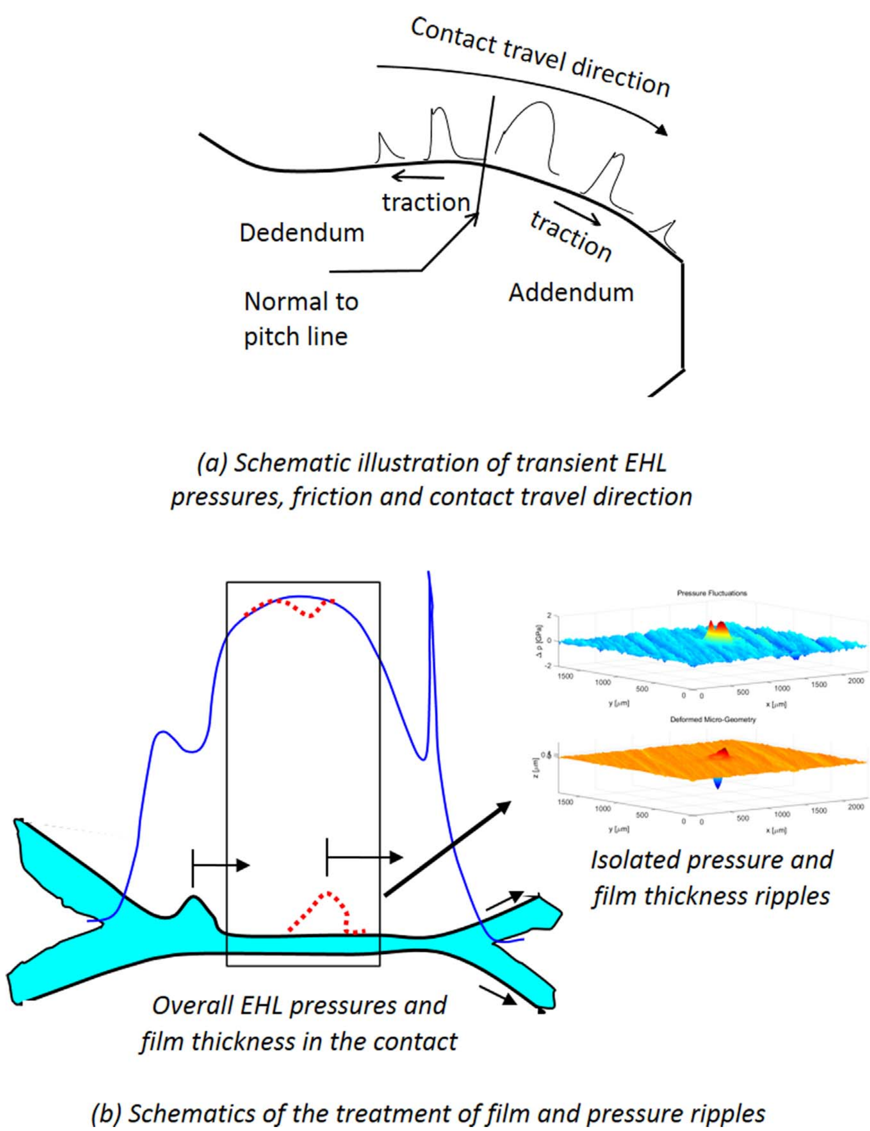
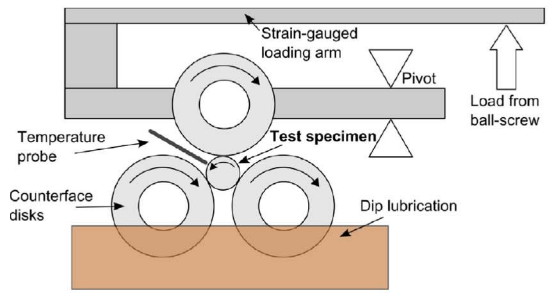
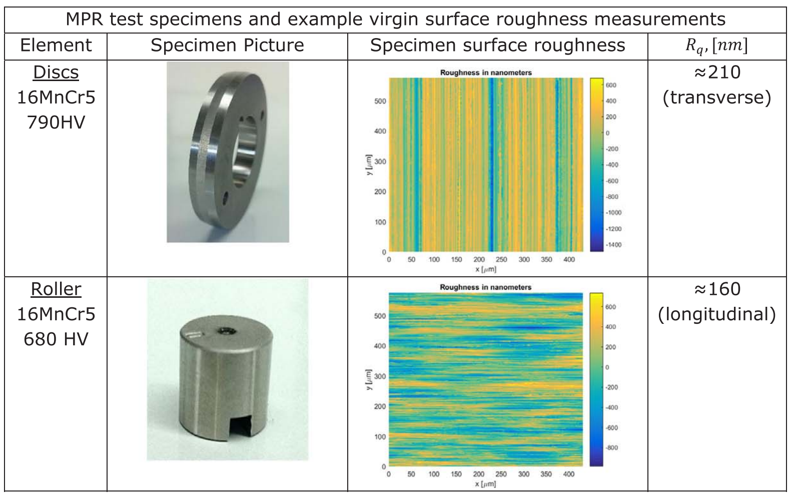


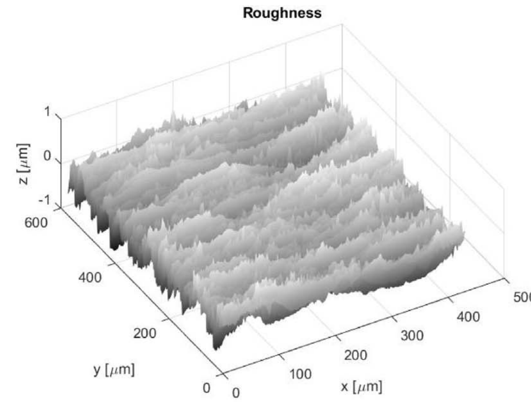
Fig. 2. (a) Schematic illustration of transient EHL pressures, friction and contact travel direction along a tooth profile during the meshing cycle. (b) Schematics of the treatment of film and pressure ripples generated by a surface disturbance (roughness) when passing through a heavily loaded lubricated contact. The red broken lines represent pressure and film ripples moving through the contact after a time $t = \Delta t$. The inlet of the contact is always on the left hand side.



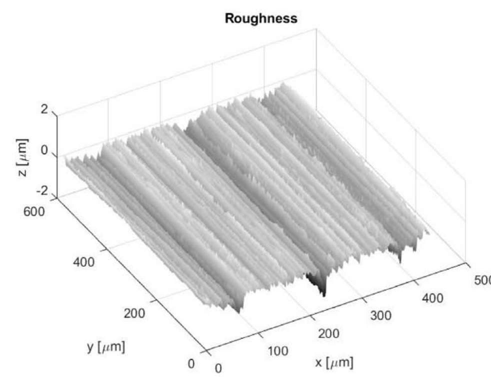
(a). Schematic illustration of the PCS Instruments triple-disc Micropitting Test Rig (MPR).



(b). Images and measured virgin topographies of an example MPR roller sample and the counterface disc.



(c) Sample 3D roughness - roller



(d) Sample 3D roughness - disc

Fig. 3. (a) Schematic illustration of the PCS Instruments triple-disc Micropitting Test Rig (MPR) used in the current micropitting experiments. (b) Images and measured virgin topographies of an example MPR roller sample and the counterface disc used in the micropitting experiments (c) Sample of the measured original surface topographies of roller and (d) discs used in the experiments and as an input to the associated micropitting prediction (Average R_q values for specimens are quoted in Fig. 3b).

the test, ensuring that micro-stress cycles are accumulated on the softer roller and hence that the micropitting damage is localized on the roller specimen. This is in line with previous micropitting tests conducted on this rig [6–8]. Both the counterface discs and roller specimen surfaces were ground, and $R_q$surface roughness values, measured using a white light optical interferometer, are approximately 210 nm and 160 nm respectively. The initial roughness of the softer test roller itself is relatively insignificant in this set-up since it is quickly changed once the micropitting starts. The roughness, and hardness, of the counterface discs on the other hand is of crucial importance in terms of reliable reproduction of micropitting. Therefore, the direction of the roughness lay for the hard counterface discs was transverse to rolling mimicking the roughness structure commonly found on gear teeth. Fig. 3b, Fig. 3c and Fig. 3d show examples of surface topographies of a roller specimen and a counterface disc obtained using an optical profilometer. The lubricant used in the tests was a Group I 15 VG oil with zinc dialkyldithiophosphate (ZDDP) anti-wear additive blended in at a treat rate of 0.1% wt phosphorus in order to minimize the wear of counterface roughness during the test.

The test operating conditions are shown in Table 1. Three slide/roll ratios, defined as sliding speed divided by entrainment speed ($S = 2(u_2-u_1) / (u_1+u_2)$) were selected: 0.05, 0.15 and 0.3 in order to simulate the conditions at various points in a spur gear pair mesh. Tests were performed at a fixed maximum Hertzian contact pressure of 1.1 GPa and a Λ-ratio of around 0.18 (calculated as the ratio of the central film thickness, predicted using the Hamrock-Dowson equation, and the measured R_q of the counterface discs only). In order to ensure that the Λ-ratio remained constant for all slide-roll ratios tested, the oil sump temperature was adjusted to give equal contact inlet temperature in each case, based on the frictional energy input in the contact and a thermal resistance network model of the MPR test rig following the procedure outlined in [23].

3. Results

3.1. Experimental results and comparison with model predictions

The surface of the MPR roller specimen was inspected at 6 and 10 million contact cycles in each test. Images of the surfaces were taken using an optical microscope and surface profiles measured with white light optical interferometer. Images of the damaged surfaces for all three slide-roll ratios after 6 million and 10 million cycles are shown in Fig. 4. Fig. 5 shows examples of circumferential sections (in rolling direction) of the damaged specimens. The section reveals presence of multiple short cracks, 10–50 μm in length, and correspondingly shallow pits, with all the characteristics generally accepted to be representative of typical micropitting damage, thus confirming that the generated surface damage is indeed that of micropitting. The surface area affected by micropitting was quantified by analysing the surface roughness

Table 1
Test conditions for the micropitting experiments conducted on the triple-disc rig.

S [-]	Disc R_q [nm]	p_o [GPa]	Entrainment speed, $\bar{u}$ [m/s]	Central film thickness, h_c [nm]	T inlet [°C]	Λ [-]
−0.05	~210	1.10	1.10	37	94	0.18
−0.15	~210	1.10	1.10	37	94	0.18
−0.30	~210	1.10	1.10	37	94	0.18

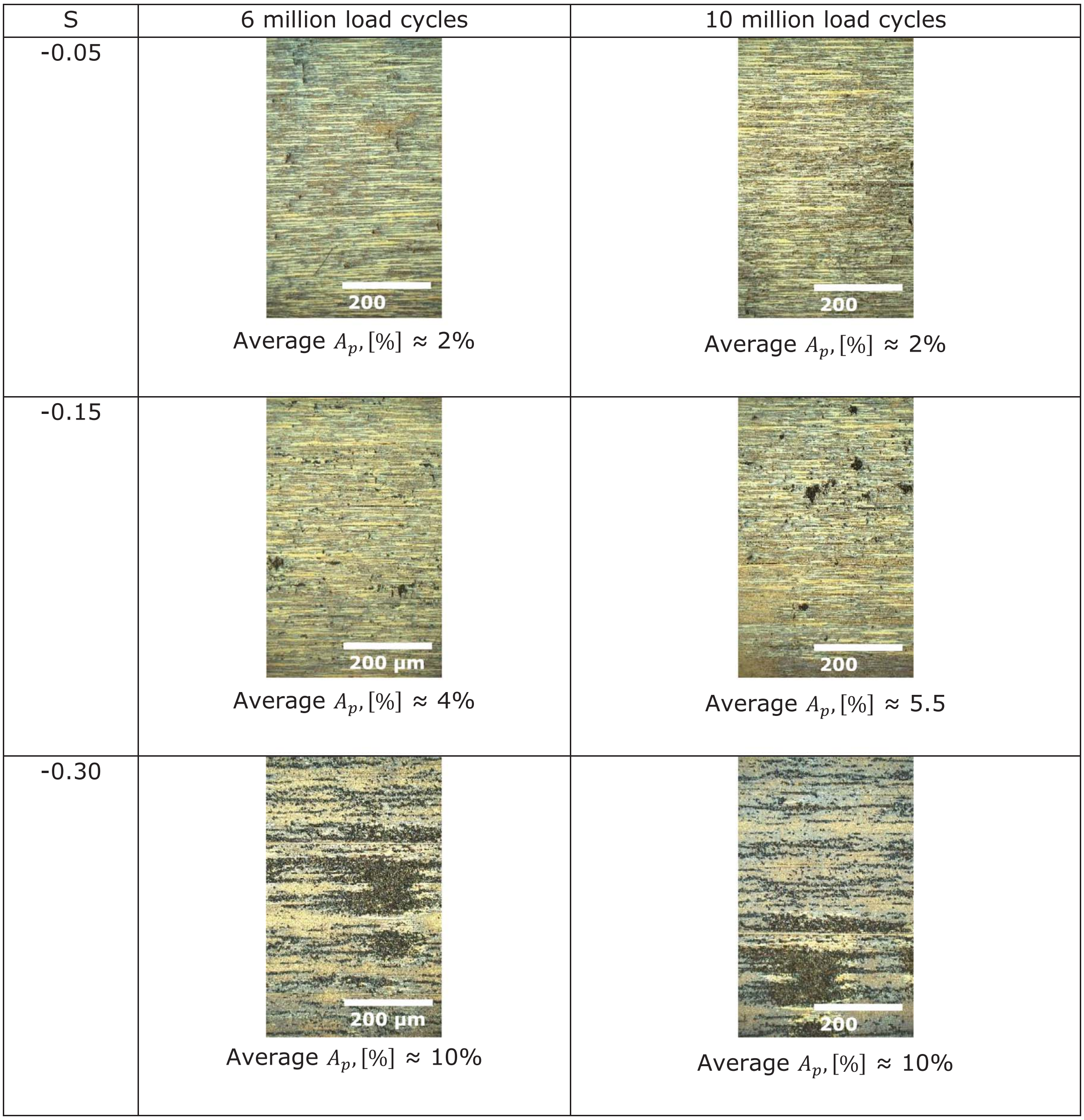


Fig. 4. Images of the micropitted roller surfaces from MPR experiments at 6 million and 10 million cycles and slide-roll ratios, S, of −0.05, −0.15 and −0.3. The measured micropitted area percentage for each case is quoted as A_p next to the corresponding image. Experimental conditions are listed in Table 1.

measurements obtained on the interferometer using a simple, in-house developed software which identifies the micropits in the measured 3D surface profiles. The results of this analysis for each test specimen are given next to the relevant image in Fig. 4 as A_p, the value which represents the fraction of micropitted area over the whole measured area. It is evident that higher magnitude of sliding produces more micropitting: the micropitted area after 10 million cycles for S = 0.3 is about 10%, for S = 0.15 it is about 5% while at S = 0.05 it is only 2%. Potential reasons for this are discussed later. It also appears that there is little difference in the amount of damage recorded at 6 and 10 million cycles, with only the case of S = 0.15 showing a discernable increase.

In order to compare these experimental results with the output of the micropitting model a simulation was performed for the same conditions over both 6 million and 10 million contact cycles as measured in the experiments. Surface topography updates were performed every 1 million cycles in these calculations. Comparison was made for each of the slide-roll ratios tested, namely S = 0.05, 0.15 and 0.3. The exact experimental conditions outlined in Table 1 and real surface topography profiles of the counterface discs and roller specimens measured before testing, as shown in Fig. 3c and Fig. 3d, were used as inputs for the simulations. Under the current experimental conditions, negligible levels of wear were recorded on both the counterface discs and roller specimens. Therefore, the wear coefficient in the model was set to 0 for these simulations.

The resulting predictions for accumulated Palmgren-Miner risk maps for the roller surface are depicted in Fig. 6. The predicted values of the overall micropitted area, A_p, for the case with S = 0.05 are 3.7% and 4.1% over 6 million and 10 million cycles respectively, for the case with S = 0.15 they are 8.1% and 9.5%, and for the case with S = 0.3 they are 9.2% and 11.3 for 6 million and 10 million cycles respectively. This can be compared with the corresponding measured experimental values shown in Fig. 4, where for S = 0.05 A_p values were about 2.0% for both 6 million and 10 million cycles, for S = 0.15 they were 4.0% and 5.5% respectively, and for S = 0.3 they were about 10.0% in both cases. In terms of absolute values, the match for the higher sliding case of S = 0.3 is excellent while the model somewhat over-estimates the amount of micropitting experienced with the lower sliding cases. The model predicts that micropitting damage continues to increase between

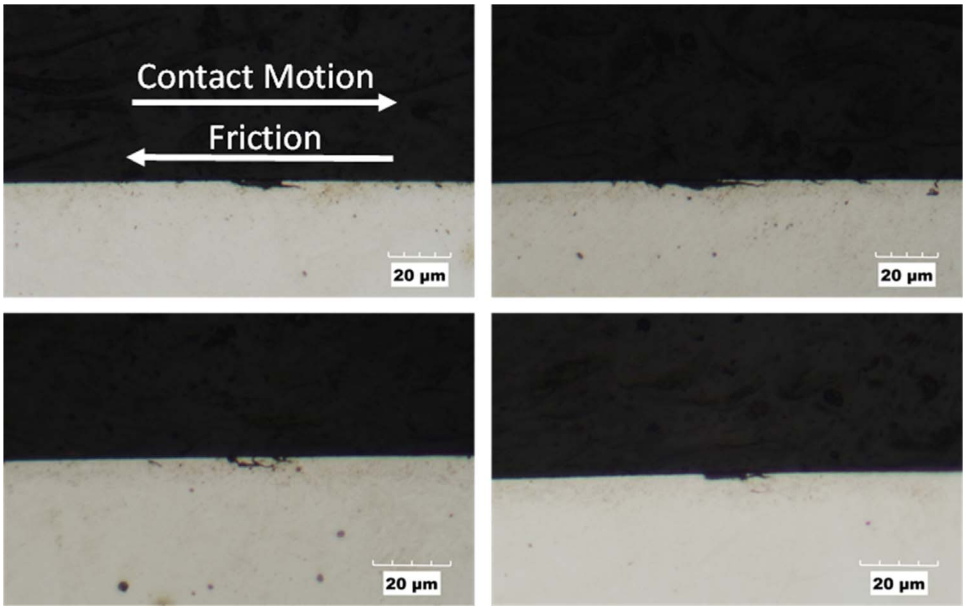


Fig. 5. Example optical images of cross sections (in rolling direction) of the roller from the S = −0.15 experiment after 10 million cycles, showing surface breaking micro-cracks and small, shallow micro-pits. Corresponding surface images are shown in Fig. 4.

6 and 10 million test cycles, all be it relatively slowly, whereas the experimental observations show that this increase is minimal. This discrepancy is most likely to be caused by the fact that some very mild wear is present in the experiments whereas the model assumes zero wear rate. Effectively, by the time the 6 million test cycles are reached even such low wear rate is able to wear the asperity peaks to the point where further micropitting damage is limited; in contrast, the assumed zero wear rate in the model, means that asperities persist even after 6 million cycles so that some progression of micropitting damage between 6 and 10 million cycles is predicted. Given the multitude of factors at play, particularly the potential unaccounted effects of wear described above but also the inevitable variations in surface roughness over the specimen surface area and the fatigue nature of micropitting damage, the predictions of the micropitting model can be considered to be in a relatively good agreement with the experimental observations, at least qualitatively. Most importantly, it is clear that in all cases the model predicts the correct relative trends in terms of increased amount of micropitting with increasing sliding. This comparison of model predictions with experimentally obtained results provides confidence that the present micropitting model can be used for prediction of micropitting under conditions pertinent to gear teeth contacts.

3.2. Prediction of micropitting damage in typical gear teeth contacts

The micropitting model was next applied to studying the occurrence of micropitting in an example pair of meshing involute spur gears. The chosen gear geometry and operating conditions used in this simulation are summarized in Table 2. The tooth geometries considered in this part of the study are assumed to have tip relief that is designed for smooth meshing, no crowning in lead direction and zero profile shifts. The gear tooth roughness used as input to the model was that measured on a commercially available spur gear with 5 mm module, 18 teeth and 50 mm facewidth. The measurements were performed on two different teeth of a single gear, and the individual profiles so measured were then assigned to the two spur gear teeth simulated in the present study. The same roughness profile was used for all calculation positions on the gear flank, i.e. it was assumed that roughness did not vary along the gear tooth flanks. The measured three-dimensional surface topographies used in the simulation are shown in Fig. 7.

Fig. 8a shows the distributions of tooth load, the associated maximum Hertz pressures and the slide-roll ratio for the driver gear during a mesh cycle. It is evident that for the conditions considered, the maximum Hertz pressure is about 1.1 GPa, occurring at the first point of single tooth pair region as expected, and the slide roll ratio is anti-symmetric around the pitch line varying between −0.74 and +0.74 at the two extremes of the path of contact.

The primary objective of this study was to examine the distribution of micropitting along the tooth flank for given operating conditions and gear micro and macro geometry. Consequently, the micropitting simulations were run at 11 different locations along the path of contact of the meshing gears, 5 on either side of the pitch line and one on the pitch line itself. The local contact conditions, including contact pressures and entrainment and sliding velocities, at the selected locations along the path of contact were first calculated using a standard approach for path of contact calculations as employed in [30] for example. The calculated contact conditions at the 11 selected locations during a mesh cycle, specified in terms of rolling angle, are summarized in Table 3.

All gear micropitting simulations in this section are performed over 3 million cycles. At the specified gear speed of 957 rpm, this corresponds to roughly the first 50 h of operation of the gear pair so that the results are representative of the development of micropitting relatively early in gear life. The evolving surface roughness profiles are updated at intervals of 1 million cycles.

3.2.1. Distribution of micropitting along the tooth flank and the effect of wear. The micropitting model was run for each of the locations identified in Table 3. The analyses were carried out using three different values of dimensional wear coefficient ($k_{lub} = 1 \times 10^{11}$ [s] and $k_{lub} = 5 \times 10^{11}$ [s]) as well as the case with no wear ($k_{lub} = 0$) in order to demonstrate how the severity of mild wear affects micropitting damage. The results for both the driving and driven gears are summarized in Table 4 where the predicted extent of micropitting at each location is expressed as A_p, a ratio of calculated micropitted area to the total analysed surface area, in the same manner as for the test cases presented earlier.

The calculated values of micropitted areas from Table 4, together with pressure and slide-roll ratio distributions, are plotted against the roll angle in Fig. 8b and c for both the driver and driven gear. To complement these results, Fig. 9 shows maps of predicted micropitted damage for the 5 selected locations in the dedendum of the driver gear to serve as an example of model output; the addendum and the corresponding locations on the driven gear show similar maps. Finally, the present simulations account for the transient nature of stresses at any given location on the tooth flank as tooth surfaces slide/roll past one another and hence the progression of calculations is best illustrated via

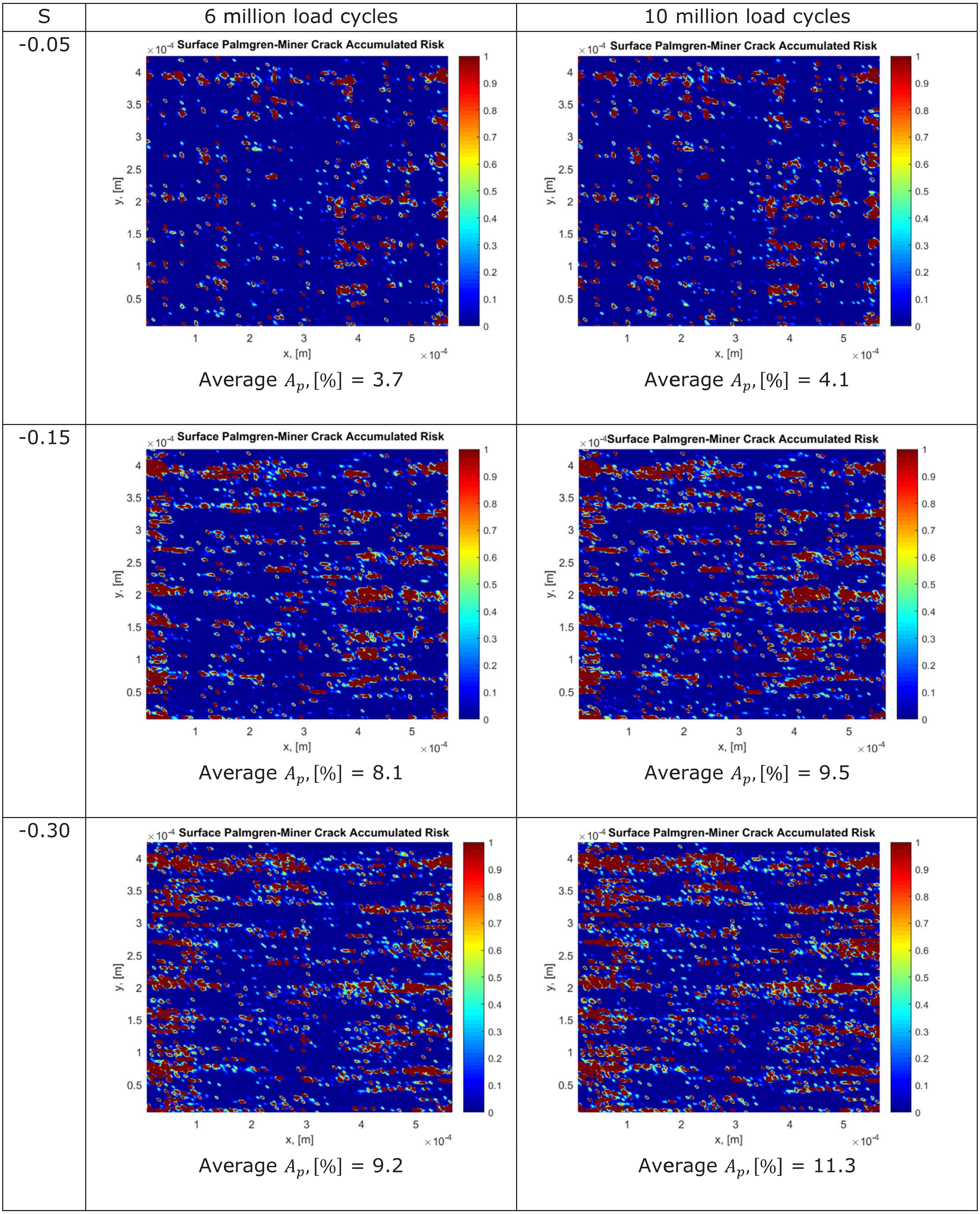
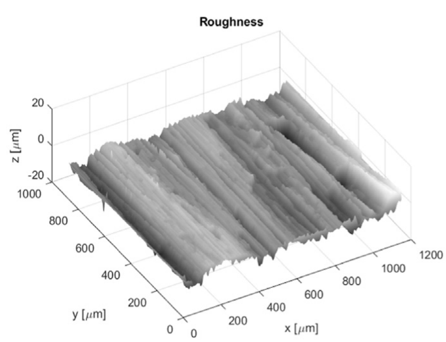


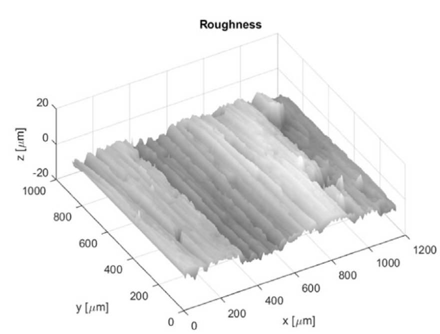
Fig. 6. Maps of damage risk on the MPR roller specimens as predicted by the current model for the experimental conditions specified in Table 1 with assumed zero wear rate. Equivalent experimental results are shown in Fig. 4. Dark red areas indicate the predicted presence of micropits.

Table 2
Spur gear geometry, operating conditions and oil properties used in the predictions of micropitting damage in gear teeth contacts (pinion and wheel are identical).

Gear Parameter	Value	Units
Module	3.95	[mm]
Gear ratio	1	[-]
Number of teeth	23	[-]
Reference pressure angle	25	[degrees]
Facewidth	19.5	[mm]
Reference diameter	90.850	[mm]
Addendum modification coefficient	0.0	[-]
Pinion/wheel tip diameter	98.750	[mm]
Pinion/wheel root diameter	80.975	[mm]
Base diameter	82.338	[mm]
Total radial backlash	0.165*module	[-]
Centre Distance (not 'tight-mesh')	91.502	[mm]
Contact ratio	1.298	[-]
Torque transmitted	251	[Nm]
Rotational speed	957	[rpm]
Oil viscosity at working temperature	90	[mPa s]
Oil piezo-viscosity exponent at working temperature, α	21	[1/GPa]
Eyring stress of lubricant	9	[MPa]
R_q of combined surfaces	4.09	[μm]
Number of load cycles simulated, N	3	[Million]



(a) Representative surface sample of tooth flank for the driver gear, Rq=2.549μm



(b) Representative surface sample of tooth flank for the driven gear, Rq=3.215 μm

Fig. 7. Surface roughnesses assigned to the two spur gears considered in the present micropitting simulation. Shown roughness is as measured on a commercial off-the-shelf gear in two different locations. Rolling direction in simulations is along x direction i.e. transverse to roughness lay.

an animation over time; since this is not possible here, for illustration purposes, Fig. 10 shows intermediate results of the simulation for the location 3 of Table 3 at a single time step. Notice that the intermediate results shown correspond to the time step when the roughness sample is located directly in the centre of the Hertzian zone (maximum mean pressure).

Before detailed analysis of the results in these figures, it is worth clarifying two features apparent in the plots of Fig. 8. Firstly, it is evident that for the gear geometry considered here, with zero profile shift and tip relief designed for smooth meshing, the severity of micropitting damage, A_p, for both the driver and driven gears is predicted to be more or less symmetrically distributed around the pitch line i.e. similar amount of micropitting is predicted for the equivalent locations in the addendum and dedendum of each gear. The amount of micropitting on the pitch line itself is predicted to be zero. Given that the amount of sliding on the pitch line is zero and increases in the dedendum and addendum, being symmetrically distributed about pitch line, as shown in Fig. 8b and 8c, this result is in line with the assertion put forward by other micropitting studies [9,23] amongst others, that the risk of micropitting is strongly dependent on the degree of sliding in the contact. In practice, more micropitting damage is often observed in the gear dedenda [21], and this is commonly attributed to the effect of sliding direction on crack propagation, whereby negative sliding present in the dedendum as shown in Fig. 8b and 8c i.e. sliding direction is opposite to the direction of the contact movement in the dedendum, promotes crack propagation through lubricant pressurisation in the crack as discussed earlier; since the present model does not consider crack propagation but is solely based on crack initiation, accounted for through the damage criterion, it does not account for the effect of sliding direction on micropitting damage. Any small differences in the extent of predicted micropitting damage in the equivalent locations in the addendum and dedendum, where absolute amount of sliding and contact pressures are the same, are due to different roughness profiles of the driver and driven gear and hence slightly different asperity stress histories depending on the relative speeds of the two surfaces. Secondly, comparison of Fig. 8b and 8c reveals that the extent of micropitting on the driver and the driven gear for any given mesh location is predicted to be very similar. This is to be expected since the same material properties were used in calculations for both gears and both tooth profiles have no profile shift and tip relief designed for smooth meshing. Once again, any small differences in the predicted amount of micropitting on the driver and driven gear at the same rolling angle position are due to the fact that the two gears were assigned different roughness profiles and hence can be expected to have slightly different stress histories for the same location in the mesh cycle, despite the fact that both experience the same macro contact conditions at a given location.

The most striking feature in Fig. 8b and 8c is the obvious difference in the amount and distribution of micropitting damage along the tooth flank for the different severity of wear coefficients studied here. The case with no wear (k_{lub} = 0) produces significantly more micropitting than both cases where some wear was present. The case with the highest wear coefficient, $k_{lub} = 5 \times 10^{-11}$ [s], produces the lowest amount of micropitting. This result is in agreement with published experimental results [7,18] and the generally accepted explanation is that the two damage mechanisms are in competition: presence of wear helps reduce the overall roughness levels, and in particular wears off the asperity peaks through the process of running-in, and hence reduces the level of asperity micro-stresses as well as the number of micro-stress cycles experienced by a given surface point, both of which lead to reduced micropitting.

Considering the distribution of micropitting damage along the tooth flank, in case of zero wear the micropitting amount increases with increasing distance away from pitch line, with the rate of increase slowing significantly past the extremes of the single tooth pair contact region, where pressure decreases but amount of sliding is still increasing. In contrast, when wear is present, as may be the case in real gear applications, the distribution of micropitting along the tooth flank exhibits a maximum at some location between the pitch line and the tooth tip. For the case of relatively low wear, $k_{lub} = 1.10^{-11}$ [s], this maximum occurs at a location that is well within the single tooth pair contact region. This may at first appear non-intuitive since the contact conditions at this point, in terms of sliding magnitude and contact pressure, are not as severe as elsewhere on the flank. Indeed, given the strong dependence of micropitting on the contact pressure and the level of sliding, maximum micropitting may have been expected to occur

Table 3
Tooth locations (and associated contact conditions) selected for the analysis of micropitting damage (data for the driver gear; gear geometry specified in Table 2).

Position	Rolling angle [deg]	Radial position of tooth centre [mm]	Reduced contact radius [mm]	Normal contact load [N/m]	Hertz max. pressure [GPa]	Hertz semi-width [μm]	Slide- Roll ratio [-]	h_c [μm]	Λ, [-]
1	18.25	43.04	8.80	23708	0.424	35.57	−0.686	0.922	0.225
2	21.10	43.72	9.40	130393	0.963	86.22	−0.480	0.760	0.186
3	22.38	44.05	9.60	174391	1.102	100.76	−0.389	0.738	0.180
4	24.92	44.77	9.87	174391	1.087	102.17	−0.206	0.747	0.183
5	27.14	45.45	9.97	174391	1.081	102.69	−0.046	0.750	0.183
6 (pitch line)	27.77	45.65	9.98	174391	1.081	102.71	0.000	0.750	0.183
7	28.41	45.86	9.97	174391	1.081	102.69	0.046	0.750	0.183
8	30.63	46.61	9.87	174391	1.087	102.17	0.206	0.747	0.183
9	33.17	47.52	9.60	174391	1.102	100.76	0.389	0.738	0.180
10	34.44	48.00	9.40	130393	0.963	86.22	0.480	0.760	0.186
11	37.30	49.11	8.80	23708	0.424	35.57	0.686	0.922	0.225

near the highest and lowest point of single pair contact (HPSPC and LPSPC), where sliding speed is higher than anywhere else within the single pair contact region and the pressure is at its maximum (see Fig. 8a). However, in addition, micropitting is strongly linked to the amount of wear occurring at any given location, with two damage mechanisms competing against each other. It is this combined effect of wear, slide-roll ratio and pressure that causes the micropitting to be maximum at a point within the single pair contact region, rather than near the extremes of the tooth flank as was predicted for the case with no wear. Indeed, the inspection of results for the case of higher wear, $k_{lub} = 5.10^{-11}$ [s], reveals that in this case the point of maximum micropitting is predicted to be near the extremes of the single tooth contact region. Nearer the tip and root of each gear, i.e. areas within the two pair contact region, the contact pressure is significantly lower and hence the predicted severity of micropitting reduces in these areas, despite the fact that the slide-roll ratio is maximum in this region. The mechanisms leading to the micropitting distribution depicted here are discussed in more detail in the Discussion section of the paper.

The symmetry of micropitting about the pitch line and the similarities between the driven and driver gear predicted here, should not be generalized; for other gear pairs, where various profile modifications may be present and stress concentrations may exist due to manufacturing and mounting errors, the distribution of micropitting may not be symmetrical about the pitch line and the amount of micropitting on the driver and driven gear may be different. Provided that the contact pressure distribution can be calculated at the desired position on the tooth flank, the model is capable of dealing with influence of these factors on micropitting, but these are outside the scope of the current paper. Also, the micropitting distribution presented here is that which may exist after the first 3 million contact cycles considered here, which corresponds to about 50 h at the start of the gear pair life, and the details of the distribution may change later in the gear pair life.

3.2.2. Relative effects of contact pressure and sliding on micropitting distribution. The results presented above illustrate the combined effects of sliding magnitude, contact pressure and film thickness on the interaction between wear and micropitting. Given that all three influencing parameters, and particularly the contact pressure and sliding magnitude, vary along the tooth flank during gear meshing, it is rather difficult to establish their individual effects on the extent of micropitting in real gears. In order to isolate the effect of (macro) contact pressure a *hypothetical case* of meshing gears with geometry described in Table 2 was simulated where the slide/roll ratio was artificially fixed at ± 0.044 at all times during the mesh. Given that film thickness varies only slightly along the contact path, this approach allows for the effect of varying contact pressure to be more easily discerned. The simulation was performed only with the mild wear case with $k_{lub} = 1 \times 10^{-11}$ [s] and results are summarized in Table 5. The results can be compared with those in Table 4 which relates to the same pair of gears but with the correct (varying) sliding velocity during meshing. It is evident that the percentage of micropitted area in this hypothetical case follows the same trend as the maximum Hertzian pressure, with a similar amount of micropitting predicted at all locations within the single pair contact region where pressure is relatively constant, and much lower damage predicted in the double pair contact region where pressure is lower. With fixed slide-roll ratio the number of asperity micro-cycles everywhere on the tooth flank is the same, and since film thickness varies very little along the path of contact, micropitting may be expected to be higher in the areas of higher macro contact pressures, given its fatigue origins. Comparing this result to that in Table 4 and Fig. 8b and 8c, which show a very different distribution of micropitting along the tooth flank, helps to further illustrate the significance of the sliding magnitude on micropitting damage through its effect on the number of asperity stress cycles.

Table 4
Predicted micropitting damage at mesh positions described in Table 3 and three different wear coefficients (k_{lub}).

Position	A_p, [%] k_{lub} = 0 (driver)	A_p, [%] k_{lub} = 0 (driven)	A_p, [%] $k_{lub} = 1 \times 10^{-11}$ [s] (driver)	A_p, [%] $k_{lub} = 1 \times 10^{-11}$ [s] (driven	A_p, [%] $k_{lub} = 5 \times 10^{-11}$ [s] driver)	A_p, [%] $k_{lub} = 5 \times 10^{-11}$ [s] driven)	Location on driver: addendum /dedendum	Location on driven: addendum /dedendum
1	91	91	24	22	6.6	5.8	Ded	Add
2	89	87.6	24	21	9.0	5.1	Ded	Add
3	85	82	28	22	8.0	5.2	Ded	Add
4	71	66	36	32	8.5	5.9	Ded	Add
5	47	40	47	46	5.5	4.4	Ded	Add
6	0	0	0	0	0	0	Pitch	Pitch
7	45.9	48	45	46	5.8	5.8	Add	Ded
8	67	69	35	35	9.6	9.0	Add	Ded
9	77	81	27	25	8.5	8.4	Add	Ded
10	78	83	26	26	8.84	8.57	Add	Ded
11	80.4	87	24	24	6.6	7.15	Add	Ded

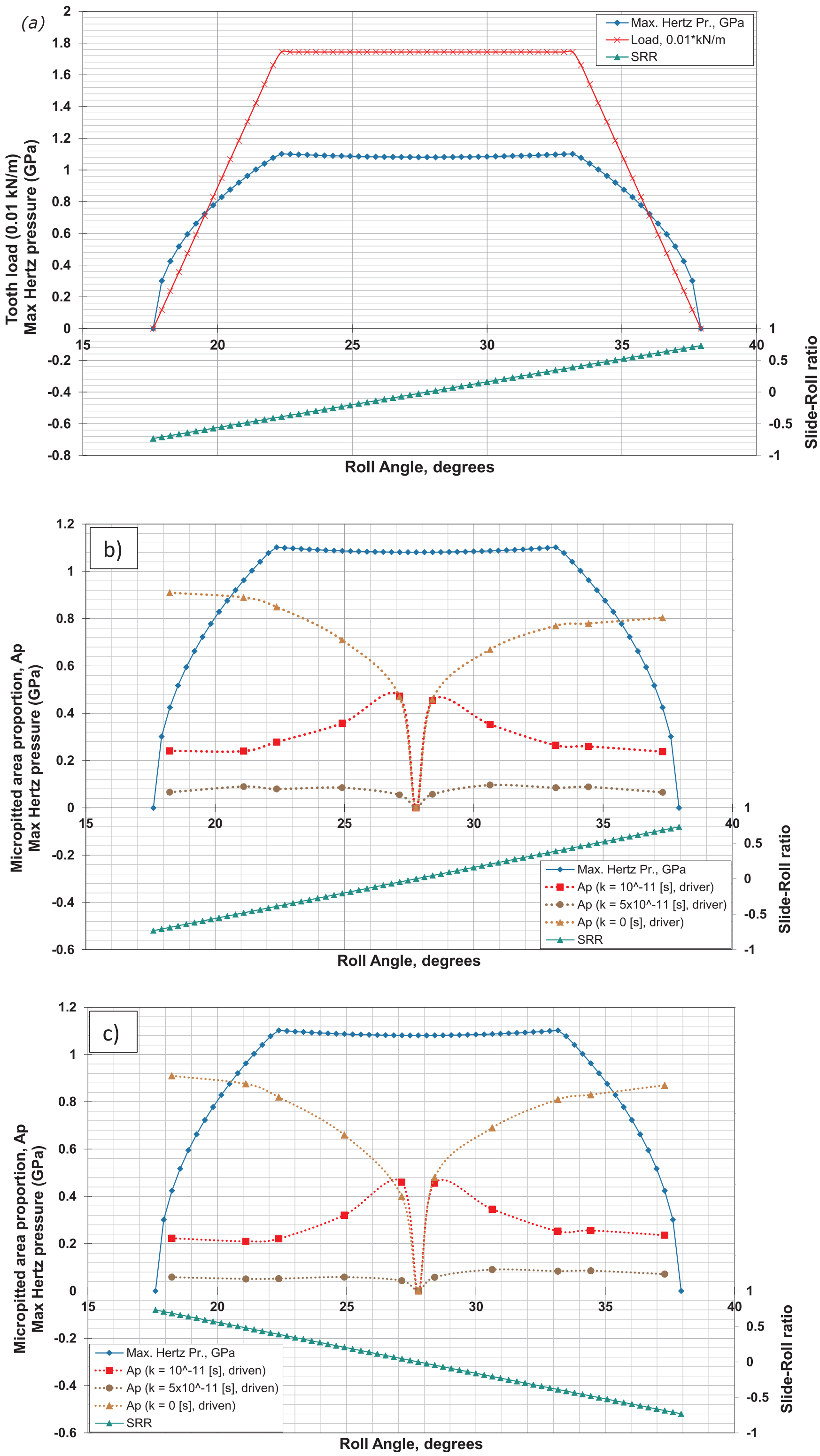


Fig. 8. (a) The tooth load, contact pressure and slide roll ratio distributions for the driver gear plotted against the roll angle during the meshing of the gear pair described in Table 2 and considered in the present micropitting simulations. Predicted micropitting distribution for three different wear coefficients ($k_{lub} = 1 \times 10^{-11}$ [s], $k_{lub} = 5 \times 10^{-11}$ [s] and $k_{lub} = 0$), slide roll ratio and maximum Hertzian pressure plotted as a function of roll angle for (a) driver gear and (b) driven gear.

3.2.3. Evaluating the effects of gear geometry on micropitting. The results above indicate that, for a given gear material, tooth roughness and lubricant, the combined effects of the local magnitude of sliding, contact pressure and film thickness determine the risk of micropitting at a particular location on the gear flank. These contact parameters are affected by the details of the gear geometry and therefore, from a design point of view, it is interesting to directly assess the influence of gear geometry parameters on micropitting propensity. The adopted methodology can be used for this purpose and, given that there is currently no universally accepted design criteria against gear micropitting, such analysis could help to provide useful guidelines to the gear designer at the initial design stage. Gear geometry is determined by a large range of parameters and it is not possible to study the influence of all of them within the scope of the current paper. The present analysis will limit itself to studying the effects of tooth number on micropitting damage, since for a given gear reference

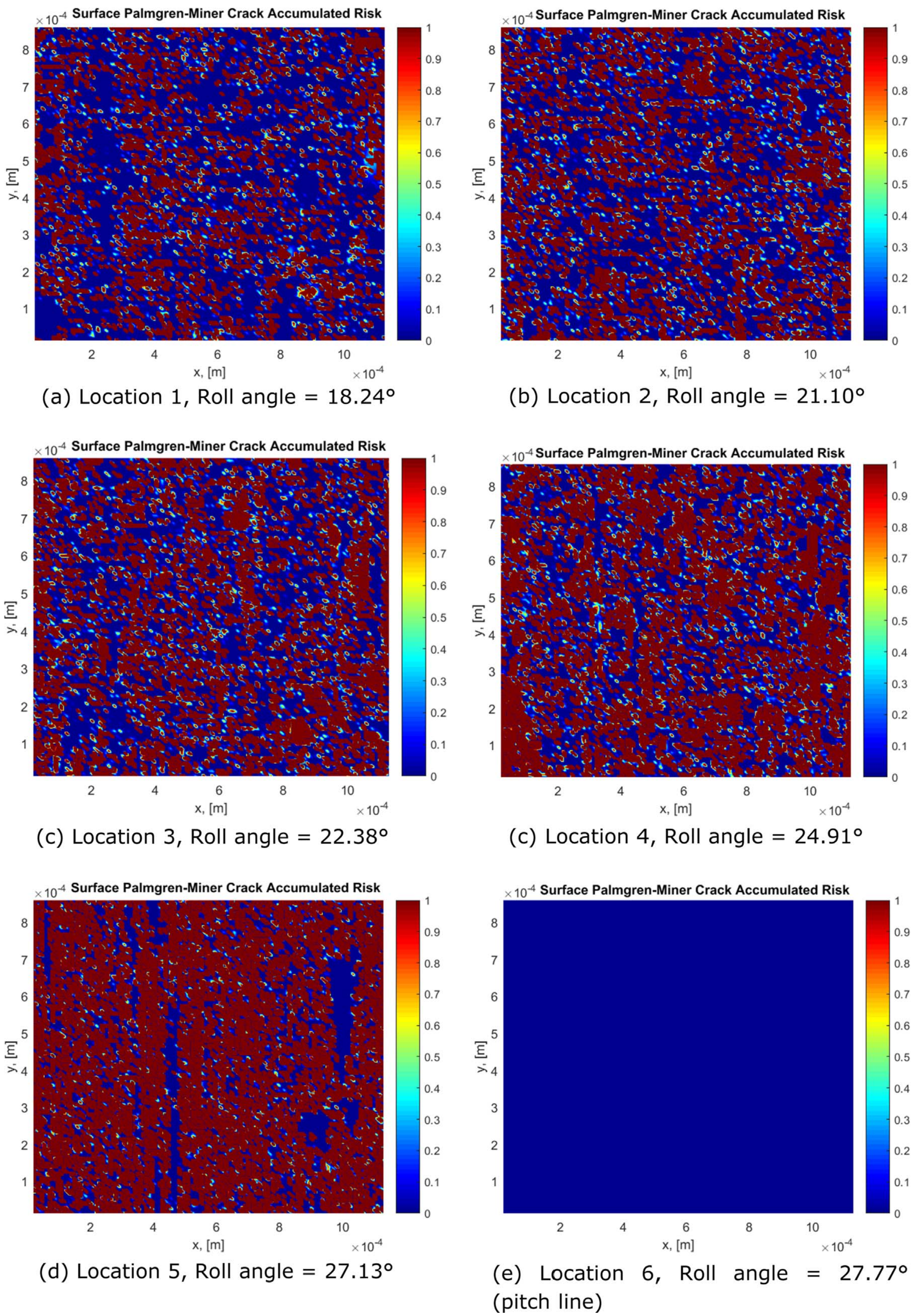
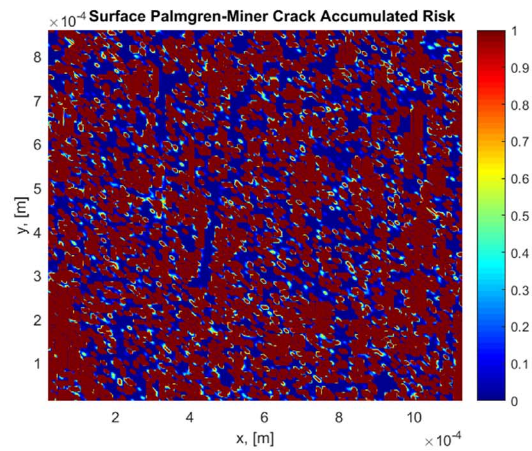
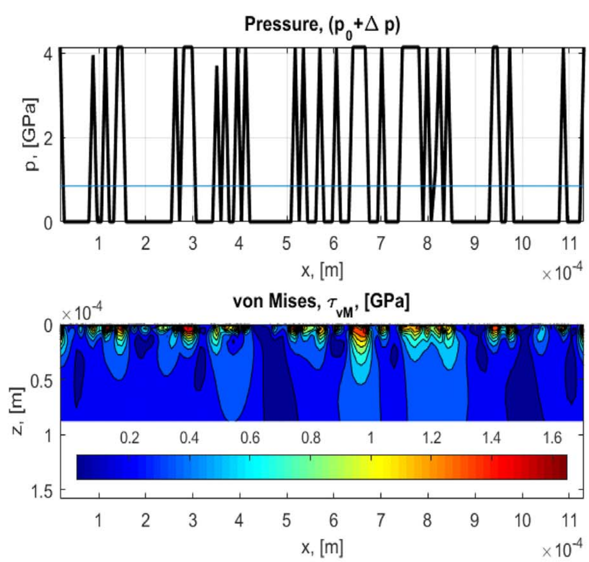


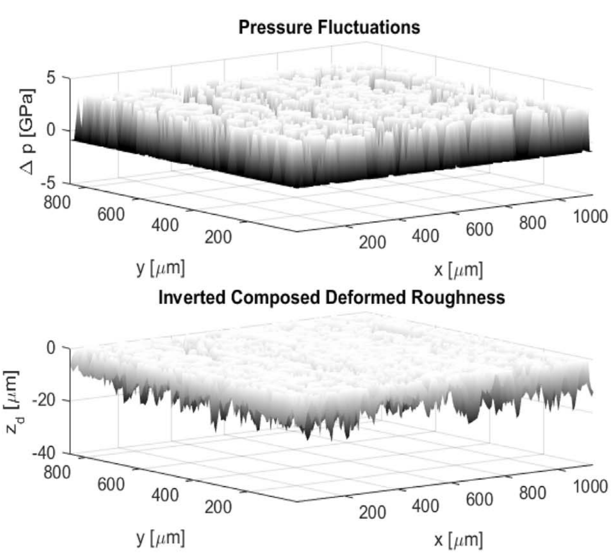
Fig. 9. Maps of micropitting damage risk predicted by the present model at selected locations in the dedendum of the driver gear for the case where the wear coefficient is $k_{lub} = 1 \times 10^{-11}$ [s]. Red areas indicate the presence of micropits. See Tables 3 and 4 for details of contact conditions at each location.



(a) Accumulated Palmgren-Miner surface crack initiation risk, (driver gear).



(b) Pressure distribution and subsurface von Mises stress contours at y=0. Time step at the centre of Hertzian zone.



(c) Pressure and surface gap at y=0. Time step at the centre of Hertzian zone.

Fig. 10. Example of intermediate results from the present micropitting model. The example shows model results for position 4 of Tables 3 and 4 and time step when moving roughness sample is in the centre of the Hertzian zone.

diameter, the choice of tooth number (and hence module) influences the amount of sliding along the contact path.

To achieve this the gears studied earlier in the paper with geometry described in Table 2, are now assigned 18 teeth, instead of the original 23 teeth, while the module is increased by a factor 23/18 so that the gear reference diameter is kept the same as before and the torque carrying capability is undiminished. The 18 tooth gears still have a zero profile shift to enable the direct comparison with the 23 tooth gear. Given the reference pressure angle of 25°, zero profile shift is acceptable for the 18 tooth gear as the undercut is avoided. The two meshing gears considered are still identical to each other in all aspects and the gear ratio remains 1. The full geometry for the meshing pair of 18 tooth gears is shown in Table 6 for the convenience of the reader.

To assess the influence of the lower tooth number on the micropitting damage, two locations in the tooth addenda on the driver gear are considered and the predicted micropitting damage for the two tooth numbers compared. Since the teeth are now of different size, some consideration needs to be given to the most appropriate way of selecting the equivalent location on the tooth flanks to enable a valid comparison. To achieve this, gear tip diameter was used as the reference position and the two analysis locations selected to correspond to radial tooth centre positions that are 0.25 x module and 0.7 x module below the tip diameter for both gears. This ensures that the comparison is made at equivalent positions on the two teeth despite the different tooth sizes. Table 7 shows the local pressures, slide-roll ratios and oil film thicknesses for the two location on the z = 18 gear as well as the equivalent data for the original z = 23 gear for comparison. Table 8 shows the predicted extent of micropitting at the two selected locations for each of the two gears with the two of the wear coefficients used previously ($k_{lub} = 0[s], 1x10^{-11}[s]$).

Comparing the contact conditions between the two gears at the location where the tooth centre is 0.7 x module radially below the tooth tip (data in rows 1 and 3 in Table 7), it is evident that both gears are within the single tooth pair region and that the contact pressure and Λ ratios are almost the same in the two cases, but the magnitude of sliding is significantly larger for the 18 teeth gear, S = 0.35 compared to S = 0.2 for the 23 teeth gear. Comparing the predicted micropitting extent for these two locations (rows 1 and 3 in Table 8), it is evident that when the wear rate is zero, more micropitting is predicted on the 18 tooth gear (A_p = 88%) than the 23 tooth gear (A_p = 67%). Given that all contact conditions other than slide-roll ratio were almost the same for the two gears at this location, the increase in micropitting damage is due to the increased magnitude of sliding in the 18 tooth gear. In contrast, when mild wear ($k_{lub} = 1.10^{-11}$ [s]) is present, the amount of micropitting is less in the higher sliding 18 tooth gear (A_p = 28%) than the lower sliding 23 tooth gear (A_p = 35%). This difference in trends between the cases with wear and no wear is the result of the sliding magnitude influencing the competition between wear and micropitting. Comparisons of the second selected location on the tooth profile, at 0.25 x module radially below tooth tip (rows 2 and 4 in Tables 7 and 8), which is in the double pair contact region for both gears, show similar overall trend regarding the influence sliding on the competition between micropitting and wear. However, the differences in micropitting damage at this location nearer the tooth tip are somewhat less pronounced. This is likely to be due to the fact that the contact pressure for the 18 tooth gear is lower than for 23 tooth gear at this location, which is likely to mitigate the influence of increased sliding, and it further illustrates the complexities in the interaction between wear, micropitting and contact conditions at a particular location on the tooth flank.

The example presented in this section illustrates the fact that the gear geometry, in this case number of teeth, can have an important influence on the extent of micropitting damage by affecting the local contact conditions, particularly the magnitude of sliding, at a given location on the tooth flank. The amount of sliding can of course be increased by changing parameters other than the tooth numbers, for example by changing the addendum modification coefficient, and such

Table 5
Predicted micropitting distribution for the driver gear of the same gear pair as in Table 3 above but for a hypothetical case where the slide-roll ratio is manually kept constant at S = ± 0.044 along the contact path to easier discern the influence of contact pressure.

Position	Radial Position of tooth centre (driver gear) [mm]	Specific film thickness, Λ	Hertz max. pressure, [GPa]	Slide-Roll ratio, S (Set to be constant)	A_p, [%] k = 1 × 10^{-11} [s] (driver)
1	43.04	0.225	0.424	−0.044	10.04
2	43.72	0.186	0.963	−0.044	43.3
4	44.77	0.183	1.087	−0.044	45.4
5	45.45	0.183	1.081	−0.044	47.2
6 (pitch line)	45.65	0.183	1.081	~0	0
7	45.86	0.183	1.081	0.044	45.3
8	46.61	0.183	1.087	0.044	45.5
10	48.00	0.186	0.963	0.044	43.1
11	49.11	0.225	0.424	0.044	9.77

changes would lead to similar effects on the competition between wear and micropitting and hence the extent and type of the resulting tooth damage. However, it should be noted that changing a single gear parameter can have multiple influences on contact conditions so that assigning any apparent differences in micropitting damage between two teeth geometries to any single mechanism is not always simple and care should be taken when interpreting any observed changes in damage patterns in the absence of detailed analysis of contact conditions.

4. Discussion

The numerical simulations and experimental measurements presented in this paper show the effect of contact conditions on the extent of micropitting damage. Both the measurements and simulations for a single elliptical contact and the model predictions for a pair of meshing spur gears indicate that sliding has a significant influence on the way micropitting will develop. In the conducted experiments, under conditions were surface wear is negligible, higher sliding produced more micropitting. The related simulations show the same trend. Given that other contact conditions, including the lubricant film thickness, temperature and friction, were kept constant in the experiments, the reasons for the observed increase in micropitting is evidently the increase in the number of contact micro-cycles induced by the counter-face roughness at higher sliding magnitudes, which in turn promotes surface fatigue which ultimately leads to micropitting. This postulate is supported by the stress histories predicted by the model at different slide-roll ratios and is also in line with the analysis presented by Kadiric and Rycerz [23].

However, perhaps the most significant observation here is the strong influence of the prevalent wear rate on micropitting damage in gear teeth contacts, both in terms of the overall level of micropitting damage and its spatial distribution along the tooth flank. In the analyses of an example spur gear pair, the case of no wear produced the highest micropitting damage, while that of highest wear produced the lowest micropitting damage. The reason for this is that presence of surface wear acts to remove the asperity peaks which in turn reduces the number of asperity stress cycles in subsequent over-rollings, which are ultimately responsible for the initiation and progression of micropitting damage. It should be noted that, although a small amount of wear can evidently be beneficial in preventing micropitting, severe wear can of course lead to failure itself, through unacceptable loss of tooth profile, even if micropitting is prevented.

In addition to the overall level of micropitting, the prevalent wear rate is also seen to affect the actual spatial distribution of micropitting damage on the tooth flank. In gear contacts the load and contact radii change with the location of the contact point on the gear flank. Excluding wear considerations, the combined effect of these variables would be to produce zero micropitting at the pitch line, where sliding is zero, higher micropitting in the areas of high sliding and lower micropitting in the areas of low sliding. Simulations of meshing gear pair under zero wear rate indeed reveal this trend: micropitting is zero on the pitch line and increases sharply towards the extremes of single tooth pair contact, as may be expected given the increase in sliding and the relatively constant contact pressure and film thickness within this region; once the double pair contact region is entered, micropitting remains at a relatively constant level with no further increases towards the tooth extremes, the trend that reflects the falling pressure in parallel with increasing sliding magnitude towards the extremes of active profile. For the equivalent gear mesh but with a relatively high wear rate present (k_{lub} = 5.10^{-11} [s]), micropitting is significantly lower everywhere on the tooth surface and its distribution along the tooth flank is relatively flat. However, for the case of relatively mild wear rate (k_{lub} = 10^{-11} [s], in current simulations) the picture becomes much more complex. The extent of micropitting is lower than that for zero wear rate and higher than in the case of high wear rate. Its distribution along the tooth flank however is remarkably different. The maximum micropitting is now predicted to occur at a location within the single pair contact region, relatively close to the pitch line (Fig. 8). This result may at first appear counter-intuitive, but it is to be expected once the competition between surface wear and micropitting is accounted for as is the case in the current model. The low wear rate means that significant asperity wear can only occur in the areas on the tooth flank where higher sliding is present. Near the pitch line where sliding is low, there will be very little wear, and hence asperity stresses will persist throughout the simulated life of the gear. Such wear distribution would result in higher micropitting in the areas of lower sliding, due to higher number of micro-contact stress cycles over a given number of meshing cycles and hence more fatigue damage, as already discussed above. Hence, the changing contact conditions along the tooth flank affect the outcome of the competition between wear and micropitting in terms of

Table 6
Gear geometry and operating conditions used in micropitting predictions for the meshing pair of spur gears where pinion and gear have 18 teeth (instead of 23 teeth considered in Table 2). As before, the pinion and wheel are identical.

Gear Parameter	Value	Units
Module	5.0472	[mm]
Gear ratio	1	[-]
Number of teeth	18	[-]
Reference pressure angle	25	[degrees]
Facewidth	19.5	[mm]
Reference diameter	90.850	[mm]
Addendum modification coefficient	0.0	[-]
Pinion/wheel tip diameter	100.944	[mm]
Pinion/wheel root diameter	78.232	[mm]
Base diameter	82.338	[mm]
Total radial backlash	0.129*module	[-]
Centre Distance (not 'tight-mesh')	91.502	[mm]
Contact ratio	1.286	[-]
Torque transmitted	251	[Nm]

Table 7
Calculated contact parameters for the two selected analysis locations on a gear tooth of the driver gear for two different tooth numbers (all other gear parameters are those shown in Table 6).

Gear pair geometry	Location on tooth: radial distance below tooth tip (within discretisation error)	Reduced contact radius [mm]	Normal contact load [N/m]	Hertz max. press. [GPa]	Slide-Roll ratio [-]	Radial Position of tooth centre [mm]	h_c [μm]	Λ [-]
$z_1 = z_2 = 23$, $m_n = 3.95$ mm, $x_1 = x_2 = 0$	$0.7 \times m_n$	9.87	174391	1.09	0.206	46.61	0.75	0.183
$z_1 = z_2 = 23$, $m_n = 3.95$ mm, $x_1 = x_2 = 0$	$0.25 \times m_n$	9.22	94832	0.83	0.548	48.36	0.79	0.192
$z_1 = z_2 = 18$, $m_n = 5.047$ mm, $x_1 = x_2 = 0$	$\sim 0.7 \times m_n$	9.68	174391	1.10	0.347	47.26	0.74	0.181
$z_1 = z_2 = 18$, $m_n = 5.047$ mm, $x_1 = x_2 = 0$	$\sim 0.25 \times m_n$	8.67	85704	0.81	0.724	49.29	0.77	0.189

surface roughness evolution along the flank, and in turn lead to the micropitting distributions shown in Fig. 8b and 8c.

The discussion above focuses on the influence of sliding in relation to the competition between wear and micropitting. Of course it is the low specific film thickness that is ultimately responsible for both of those damage modes and for this reason it is worth considering its effects too. The effect of specific film thickness, defined as the ratio of EHL film thickness to composite surface roughness and commonly denoted as Λ, on micropitting damage has been studied before, both experimentally and numerically, see for example [7,9]. In order to illustrate its effect using the current gear micropitting model, additional simulations were conducted under the experimental test conditions shown earlier in Table 1, for the case where the slide-roll ratio was equal to 0.3. To achieve different Λ ratios, the heights of the measured roughness profiles of the specimens (shown in Fig. 3b) were scaled by multiplying all height values by a desired factor. The resulting Λ ratios are listed in Table 9, together with the predictions of micropitting damage. The results show that for zero wear rate, the micropitting damage increases with decreasing Λ ratio, as may be expected. However, when mild wear is present ($k_{lub} = 1x10^{-11}[s]$), micropitting damage increases initially as Λ starts to drop, but at very low Λ ratio it is predicted to actually decrease. The reason for this is that as Λ ratio is decreased, surface wear becomes more and more dominant, so that there may exist a 'critical' lambda value where the balance of wear and micropitting changes. The trend is illustrated schematically in Fig. 11. This observation of course needs full experimental verification, but is nevertheless included here to help illustrate the complexities of the interaction between lubrication effectiveness, wear and micropitting.

It should be noted that although the use of Λ ratio alone to represent lubrication effectiveness is very practical it excludes some significant factors that may be at play. For example, previous experimental work has showed that lubricant additives can affect micropitting behaviour through their influence on wear, boundary film thickness and friction [7,8,10,31]. It has also been shown [32,33] that the rough surface local film thickness and friction, and hence stresses, and the conditions where surface separation occurs are affected by the exact structure (height and spatial variation) of the roughness being considered i.e. the same Λ ratio does not imply the same local lubrication conditions and stresses. Therefore, the exact shape of the curves illustrated in Fig. 11 will additionally depend on the lubricant chemistry and surface roughness structure.

The current model is shown to be able to predict micropitting in rolling-sliding contacts between two meshing gear teeth but it is important to note its limitations so that any obtained predictions can be correctly interpreted. Firstly, the presented model outputs are intended to reveal trends in how contact conditions and gear geometry can affect micropitting damage; the absolute values of micropitting should not be relied upon in isolation as they can change significantly depending on the exact surface roughness profile patch that is used in the model input for any given gear, which will inevitably vary somewhat between different teeth or even different areas on the same tooth, as well as with the frequency with which the surfaces are updated. The model is based on stress history and damage accumulation parameters and therefore considers only crack initiation. Fatigue cracks, such as those in micropitting, also have a propagation phase, and the relative length in time of these two phases is affected by the imposed conditions [34]. In particular, crack propagation is strongly affected by the direction of sliding: negative sliding (cracked surface slower) i.e. the motion of the contact is opposite to the direction of sliding, is believed to accelerate crack propagation. The mechanisms responsible for this are not completely understood yet, but one of the common mechanisms quoted is that negative sliding promotes lubricant ingress into the crack, opened by friction force acting opposite to the direction of the contact movement, and then the subsequent pressurisation by the over-rolling contact increases stress intensity factors at the crack tip and hence increases crack growth rates [35,36]. This results in surface cracks propagating into the

Table 8
Predicted micropitting damage at the two selected locations on the tooth flank of the driver gear for the two gear designs with different tooth numbers and for two different wear coefficients.

Gear pair geometry	Location on tooth: radial distance below tooth tip (within discretization error)	A_p, [%] k_{lub} = 0 [s] (driver gear)	A_p, [%] $k_{lub} = 1 \times 10^{-11}$ [s] (driver gear)	Location (driver gear)
$z_1 = z_2 = 23$, $m_n = 3.95$ mm, $x_1 = x_2 = 0$	~0.7 x m_n	67	35	Addendum
$z_1 = z_2 = 23$, $m_n = 3.95$ mm, $x_1 = x_2 = 0$	~0.25 x m_n	79	24	Addendum
$z_1 = z_2 = 18$, $m_n = 5.047$ mm, $x_1 = x_2 = 0$	~0.7 x m_n	88	28	Addendum
$z_1 = z_2 = 18$, $m_n = 5.047$ mm, $x_1 = x_2 = 0$	~0.25 x m_n	88	25	Addendum

Table 9
Predicted micropitting results under the conditions used in the triple-disc experiments but now with three different simulated Λ ratios (achieved by scaling the original roller specimen roughness, which corresponds to the first row in the table, by an appropriate factor). All other contact conditions are the same as those modelled earlier and listed in Table 1.

S, [-]	p_o [GPa]	Entrainment speed, $\bar{u}$ [m/s]	Λ, [-]	A_p, [%] for $k_{lub} = 1x10^{-11}$	A_p, [%] for $k_{lub} = 0$
0.30	1.10	1.10	0.175	0.073	7.19
0.30	1.10	1.10	0.245	0.080	2.78
0.30	1.10	1.10	0.307	0.056	0.24

subsurface in the direction opposite to that of the applied friction force, as is indeed evident in the current experimental results (Fig. 5). This relationship between friction direction and crack growth direction is responsible for the fact that cracks on either side of the pitch line on a gear tooth are seen to propagate in opposite directions when tooth is sectioned. The mechanism is also quoted as the reason why in practical applications, the micropitting damage is often more pronounced in the dedendum of a gear tooth. Since the current model does not consider these effects, it is unable to account for any potential differences in damage between the dedendum and addendum due to crack propagation effects and the predicted micropitting distribution is close to symmetrical around the pitch line, any differences being caused only by the differences in surface roughness of the two mating gears. In addition, the driver and driven gears considered here were assigned very similar surface roughness (measured on two different teeth on the same gear) with almost equal R_q values. The material properties of the gears were also the same. It has been shown that in low-wear situations, when one of the surfaces in contact possesses a roughness substantially higher than the other, the smoother surface is exposed to a higher number of stress cycles if the rougher surfaces is travelling faster [5–8,37]. This can lead to damage accumulating preferentially on the slower, smoother surface, particularly if the rougher surface is also harder so that its asperities persist throughout the contact lifetime. Finally, the current model is isothermal, in that the bulk temperature of the gear teeth is considered constant throughout the simulated lifetime and therefore needs to be known as it is one of the inputs to the model. However, the inlet shear heating effects, which act to reduce EHL film thickness and can be important in gear teeth contacts, particularly at higher speeds, are included through an appropriate correction to EHL film thickness based on published literature [38].

5. Conclusion

This paper studies the occurrence of micropitting damage in gear teeth contacts through numerical simulations and experiments. Predictions of micropitting damage are achieved by appropriately adapting an existing micropitting model so that it can be applied to typical spur gear teeth contacts, including case carburized gear steels and transient contact conditions occurring during gear meshing. Importantly, the model accounts for the simultaneous effect of wear and micropitting on the evolution of surface roughness during the gear operation. The model is validated by comparing its predictions to measurements obtained on a triple-disc rolling contact fatigue rig under controlled contact conditions. The main conclusions of this study can be summarized as follows:

- The interaction between surface wear and micropitting during contact operation determines the evolution of surface roughness over time and subsequently has a very significant effect on the resulting level and type of surface damage. The two failure modes are in competition with one another and the outcome of this competition is heavily dependent on the prevailing contact conditions, particularly the level of sliding. Hence, an accurate gear micropitting model should not only account for mixed lubrication conditions and stress history resulting from the rough tooth surfaces rolling-sliding over one another under transient contact conditions during a mesh cycle, but it must also consider mild surface wear and micropitting simultaneously, as is the case with the present model.
- The presence of mild wear can reduce the amount of micropitting through removal of roughness asperity peaks on one or both of the contacting surfaces. This in turn reduces the number of asperity stress cycles and hence reduces the fatigue damage accumulation which is ultimately responsible for the onset of micropitting. Both excessive wear and/or micropitting can lead to failure, so the ideal situation is that in which some mild wear at the start of machine operation appropriately wears-in the surfaces, thus reducing subsequent micropitting, followed by zero or negligible wear for the rest of the machine life time.
- An increase in sliding speed increases the number of fatigue microcycles from asperity contacts, but it also increases wear. Hence, if no wear is present increased sliding results in increased micropitting as seen in the current experiments and predictions, but with wear present the influence of sliding on micropitting damage is more complex and ultimately depends on the outcome of the competition between wear and surface fatigue under a given slide/roll ratio.
- This complex interaction between sliding magnitude, surface wear and micropitting affects the distribution of micropitting damage along the tooth flank, since the sliding magnitude varies along the contact path: with zero wear the micropitting damage is at a maximum at or near the highest and lowest point of single tooth pair

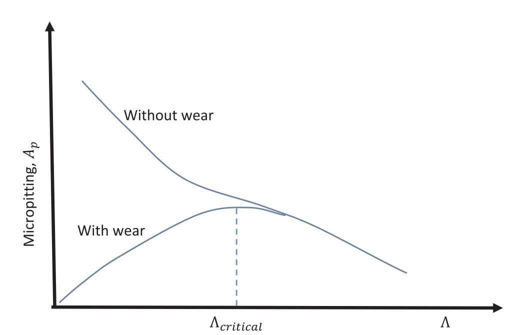


Fig. 11. A schematic representation of the expected trends in micropitting behaviour with specific film thickness, Λ for cases with and without the presence of mild wear.

contact region, as may intuitively be expected; but when mild wear is introduced, the location of maximum micropitting damage can move well inside the single pair region.

- Present model can be used to assess the impact of gear geometry on micropitting. A gear with the same diameter but fewer teeth (larger module) was shown to suffer less micropitting than an equivalent gear with more teeth. However, with mild wear, micropitting can reduce as the number of teeth is decreased. This is due to the interaction between sliding, contact pressure, wear rate and surface fatigue.
- The model only considers crack initiation, which inevitably results in micropitting damage distribution which is more or less symmetrical about the pitch line. The model is therefore unable to predict the differences in micropitting damage between dedendum and addendum which are often observed in practice and which are commonly thought to occur due the effects of the relative directions of sliding and contact movement on surface crack propagation.

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